

Chapter-5

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COORDINATION COMPOUNDS

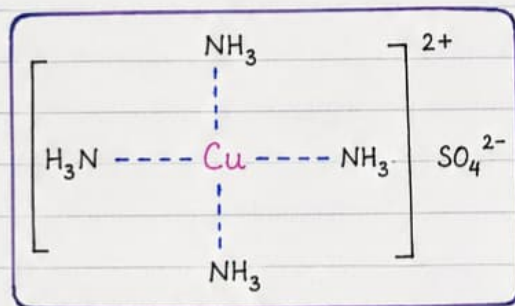
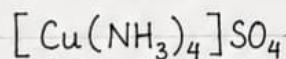
Introduction

- * Coordination compounds are those compounds which contain a **central metal atom or ion** surrounded by **ligands**.
- * These compounds are also called **complex compounds**.
- * The central metal atom/ion and the surrounding ligands together form a **coordination entity** (complex ion or molecule).

Important Terms

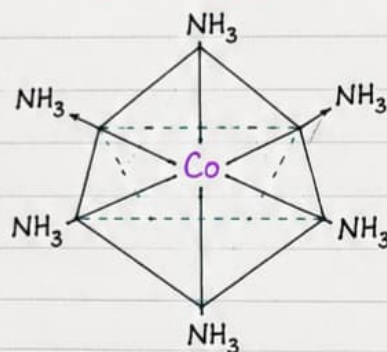
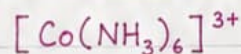
- * **Coordination Entity** : The species which exists as a unit with definite composition.
- * **Central Metal Atom/Ion** : The metal atom or ion present at the centre.
- * **Ligand** : The ions or molecules which are attached to the central metal atom/ion through **coordinate bond**.
- * **Coordination Number (C.N.)** : The number of donor atoms directly attached to the central metal atom/ion.
- * **Coordination Sphere** : The central metal ion and the ligands directly attached to it.
- * **Counter Ion** : The ions present outside the coordination sphere to balance the charge.

[Example]



Here, $[Cu(NH_3)_4]^{2+}$ is complex ion and SO_4^{2-} is counter ion.

3D Structure Example



Octahedral Geometry
(C.N. = 6)

Classification of Ligands

On the basis of Charge	On the basis of Donor atoms	On the basis of Nature
• Anionic ligands (Cl^-, CN^-, OH^-) • Cationic ligands (NH_3^+, NO^+) • Neutral ligands (NH_3, H_2O, CO)	• Monodentate (one donor atom) $\rightarrow NH_3$, H_2O, Cl^- • Bidentate (two donor atoms) $\rightarrow$ ethylenediamine (en) • Polydentate (more than two donor atoms) $\rightarrow$ EDTA (6 donor atoms)	• Aliphatic ligands (en, oxalate) • Aromatic ligands (pyridine) • Chelating ligands (en, EDTA)

Chelate Effect :

Chelating ligands form more stable complexes than similar ligands with same donor atoms.



I. Nomenclature of Coordination Compounds

1. Name of **cation** is written first, then **anion**.
2. Name of **ligand** is written next.
3. Number of ligand is shown by prefix **di-**, **tri-**, **tetra-**, **penta-**, **hexa-** etc.
4. Name of central metal ends with its **oxidation state** in **Roman numeral** in brackets.

Examples :

- $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3 \rightarrow \text{Hexaamminecobalt(III) chloride}$
- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2] \rightarrow \text{Diamminedichloridoplatinum(II)}$
- $\text{K}_3[\text{Fe}(\text{CN})_6] \rightarrow \text{Potassium hexacyanidoferrate(III)}$
- $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+ \rightarrow \text{Tetraaquadichloridochromium(III) ion}$

III. Coordinate Bond

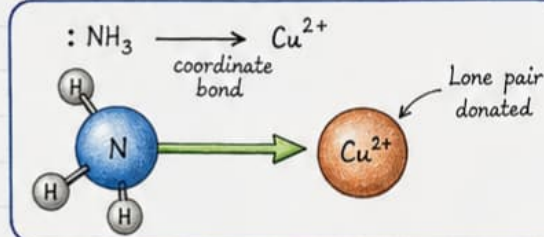
- * A coordinate bond is formed when a ligand donates a **lone pair of electrons** to the metal atom/ion.
- * These bonds are **covalent** in nature and **directional**.
- * Represented by an arrow ($\rightarrow$) from ligand to metal.

II. Types of Ligands

1. **Monodentate (unidentate)** :
Donate one pair of e^- .
e.g., NH_3 , Cl^- , H_2O
2. **Bidentate (chelating)** :
Donate two pairs of e^- .
e.g., **en** (ethylenediamine),
ox (oxalate)
3. **Polydentate** : Donate more than two pairs of e^- . e.g., **EDTA** (hexadentate)

Note :

Ligands form coordinate bond using lone pair of e^- with the central metal atom/ion.



IV. Coordination Number (C.N.) & Geometry

- * Coordination number is the number of donor atoms directly attached to the central metal atom/ion.

* Common geometries :

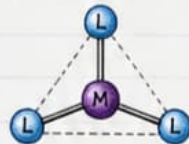
Coordination Number	Geometry	Shape (3D)	Example
2	Linear		$[\text{Ag}(\text{NH}_3)_2]^+$
4	Tetrahedral		$[\text{NiCl}_4]^{2-}$
4	Square Planar		$[\text{PtCl}_4]^{2-}$
6	Octahedral		$[\text{Co}(\text{NH}_3)_6]^{3+}$

3D Geometry Diagrams

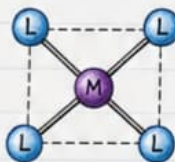
Linear (C.N. = 2)



Tetrahedral (C.N. = 4)



Square Planar (C.N. = 4)



Octahedral (C.N. = 6)



V. Isomerism in Coordination Compounds

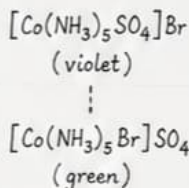
1. Structural Isomerism :

- Ionisation Isomerism • Hydrate Isomerism • Linkage Isomerism

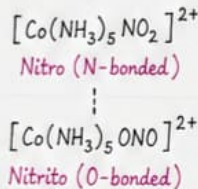
2. Stereoisomerism :

- Geometrical Isomerism • Optical Isomerism

(a) Ionisation Isomerism



(b) Linkage Isomerism



Quick Tip

Always check ligand type, coordination number and oxidation state while naming.

Coordination compounds play a vital role in **biological systems**, **catalysis** and **analytical chemistry**. ★

VI. Oxidation State of Central Metal

- * Oxidation state (O.S.) of the central metal ion is calculated by considering the charge on the complex ion and the charges on all the ligands.
- * It helps in understanding **electronic configuration** and **magnetic behaviour**.

Example: $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Let O.S. of Co = x

$$x + (6 \times 0) = +3 \quad (\text{since } \text{NH}_3 \text{ is neutral, and overall charge of cation is } +3)$$

$$\Rightarrow x = +3$$

VII. Werner's Theory

* Proposed by **A. Werner** in 1893.

* Postulates:

- (1) A metal exhibits **two types** of valencies.
Primary valency = Oxidation state (ionisable)
Secondary valency = Coordination number (non-ionisable)
- (2) Primary valencies are satisfied by anions.
- (3) Secondary valencies are satisfied by ligands.
- (4) Secondary valencies have fixed direction in space.
- (5) A coordination compound has a **definite geometry**.

Example: $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Primary valency of Co = 3 (ionisable, satisfied by 3 Cl^-)

Secondary valency of Co = 6 (non-ionisable, satisfied by 6 NH_3)

Geometry = **Octahedral**



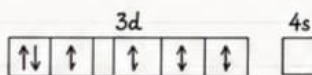
VIII. Magnetic Properties

* Depends on the number of unpaired electrons.

No. of unpaired electrons	Magnetic Behaviour	Example
0	Diamagnetic	$[\text{Zn}(\text{NH}_3)_4]^{2+}$
1 - 5	Paramagnetic	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$
5	Strongly Paramagnetic	$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$

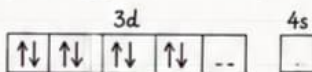
Example: Electronic configuration & magnetism

1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ ($\text{Fe}^{2+} \rightarrow d^6$)



4 unpaired $e^- \rightarrow$ Paramagnetic

2) $[\text{Fe}(\text{CN})_6]^{4-}$ ($\text{Fe}^{2+} \rightarrow d^6$, CN^- strong field ligand)



0 unpaired $e^- \rightarrow$ Diamagnetic

Note:

Weak field ligands $\rightarrow$ more unpaired electrons

Strong field ligands $\rightarrow$ less unpaired electrons



IX. Shape (Geometry) of Some Common Coordination Compounds

Coordination Number	Geometry	3D Structure	Example
2	Linear (180°)		$[\text{Ag}(\text{NH}_3)_2]^+$, $[\text{HgCl}_2]$ * Ligands opposite to each other.
4	Tetrahedral (109.5°)		$[\text{ZnCl}_4]^{2-}$, $[\text{NiCl}_4]^{2-}$ * Ligands at corners of tetrahedron.
4	Square Planar (90°)		$[\text{PtCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$ * Ligands in same plane at 90° .
6	Octahedral (90°)		$[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$ * Four ligands in plane & two above/below.

Common Ligands

- NH_3 - ammine (0)
- H_2O - aqua (0)
- Cl^- - chloro (-1)
- Br^- - bromo (-1)
- I^- - iodo (-1)
- CN^- - cyano (-1)
- OH^- - hydroxo (-1)
- NO_2^- - nitro (-1)
- NO_2 (via N) - nitro (-1)
- CO - carbonyl (0)
- en - ethylenediamine (0)
- ox^{2-} - oxalato (-2)
- EDTA^{4-} - ethylenediamine tetraacetate (-4)

★ Quick Revision

- Coordination compounds contain a central metal atom/ion surrounded by ligands.
- Key points: Nomenclature, Ligands, Coordination number, Geometry, Magnetic properties.
- Remember: Strong field ligands $\rightarrow$ Low spin (Diamagnetic), Weak field ligands $\rightarrow$ High spin (Paramagnetic).

Practice structures and colours of complexes for better retention! ♥

X. Bond Theories

1. Valence Bond Theory (VBT)

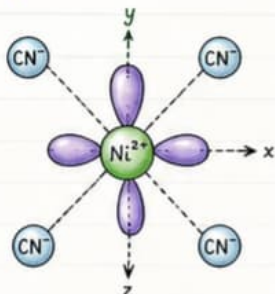
- * Central metal ion uses hybrid orbitals of suitable type to accommodate ligands.
- * It explains geometry and magnetic behaviour.
- * Generally involves ionisation of metal and pairing of electrons.

Example : $[\text{Ni}(\text{CN})_4]^{2-}$ (Square Planar)



Uses dsp^2 hybridisation.

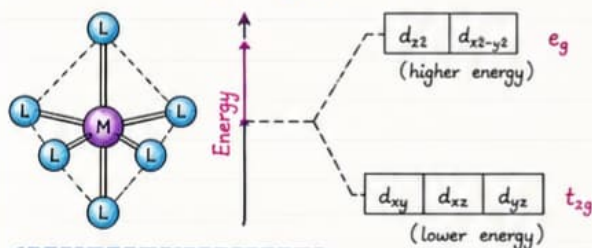
Two 3d, one 4s and two 4p orbitals hybridise to form 4 dsp^2 orbitals lying in one plane at 90° .



2. Crystal Field Theory (CFT)

- * Ligands are treated as point charges or dipoles that create an electrostatic field around the central metal ion.
- * Explains colour, magnetic properties and stability of complexes.
- * d-orbitals split into two sets of different energy.

Octahedral Splitting of d-orbitals



Δ_o = Crystal field splitting energy
Greater $\Delta_o \rightarrow$ Low spin complexes
Smaller $\Delta_o \rightarrow$ High spin complexes

XIII. Some Important Examples

Complex	Name	Colour	Geometry	C.N.
$[\text{Co}(\text{NH}_3)_6]^{3+}$	Hexaamminecobalt(III) ion	Yellow	Octahedral	6
$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	Hexaaquacopper(II) ion	Blue	Octahedral	6
$[\text{Ni}(\text{CO})_4]$	Tetracarbonyl nickel(0)	Greenish yellow	Tetrahedral	4
$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	Hexaaquachromium(III) ion	Violet	Octahedral	6
$[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	Diamminedichlorido platinum(II)	Yellow	Square Planar	4

Common Ligands with Denticity

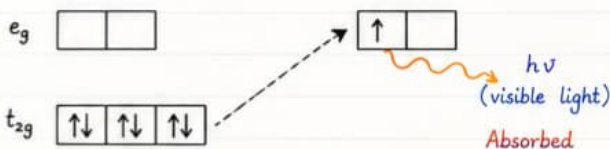
Water (H_2O) (1)	Ammonia (NH_3) (1)	Chloride (Cl^-) (1)	Cyanide (CN^-) (1)	Ethylenediamine (en) (2)	Oxalate (ox^{2-}) (2)	EDTA $^{4-}$ (EDTA) (6)
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(Number in bracket shows denticity)

XI. Colour of Coordination Compounds

- * Most coordination compounds are coloured.
- * Colour arises due to d-d electronic transitions or charge transfer transitions.

d-d Transition (involves same metal ion)



Electron jumps to higher energy level by absorbing visible light.

Charge Transfer Transition

Electron moves from ligand to metal ($L \rightarrow M$) or metal to ligand ($M \rightarrow L$) on absorbing light.

LMCT : $L \rightarrow M$

MLCT : $M \rightarrow L$

XII. Stability of Coordination Compounds

- * More the stability, less is the tendency to dissociate in solution.
- * Depends on nature of metal ion, ligands and charge.

Factors	Effect on Stability
Charge on metal ion	Higher charge $\rightarrow$ More stability
Nature of metal	Smaller size & higher charge density $\rightarrow$ More stability
Nature of ligand	Chelating > Monodentate (due to chelate effect)
Coordination number	Higher C.N. $\rightarrow$ More stability
Solvent	Low polarity $\rightarrow$ More stability

★ Key Points (Quick Revision)

1. Coordination compounds contain a central metal atom/ion with ligands attached to it.
2. Ligands donate lone pair of electrons to the metal.
3. Coordination number = No. of donor atoms attached.
4. Geometry depends on coordination number.
5. O.S., magnetic behaviour and colour help in understanding complexes.
6. Stability increases with chelation, higher charge and appropriate ligand.



Remember!

Change in ligand or central metal can change O.S., geometry, colour and magnetic behaviour.